

Ion Flight — Teacher Guide

Murray State University · ideal linear TOF mass spectrometry

WHAT STUDENTS DO

Students send labeled ion packets through a linear time-of-flight (TOF) drift tube, then read the detector's **time** trace. They adjust tube length, acceleration voltage, and named Gaussian timing width (σ), compare packets, and answer three prediction questions. It runs offline: open **index.html** in any modern browser. The animation is deliberately speed-scaled, while every displayed arrival is in real microseconds.

THE MODEL UNDERNEATH

$$K.E. = qV = \frac{1}{2}mv^2 \rightarrow t = L \sqrt{m/(2qV)}$$

The model converts Da to kg and charge number z to $q = z \cdot e$ coulombs. It starts the clock at entry to the **field-free drift region**, after acceleration. The consequential claim is modest: an ideal linear TOF separates arrival times by **m/z** . It does not identify a molecule or model the source, extraction optics, reflectron, detector response, calibration, or fragmentation.

Prediction to take before running: Which arrives first: 100 Da at $z = 1$, or 100 Da at $z = 2$? Then ask whether 200 Da at $z = 1$ can be told apart from 400 Da at $z = 2$. The first exposes charge's role; the second exposes the m/z limit.

12–15 MINUTE SESSION

- **0–2** Open the how-to. Ask students to name the source, drift tube, detector, and clock. Emphasize that this is a linear TOF cutaway.
- **2–5** Use **Same mass, different charge**. Gather a vote, run it, then point to the smaller arrival time for $z = 2$. Connect the result to qV and $\sqrt{q}$.
- **5–8** Use **Different mass, equal m/z** . Let the coincident trace challenge “a mass spectrometer measures mass” as an incomplete statement.
- **8–12** Return to the three-packet set. Make L longer with fixed absolute σ , then repeat with fixed fractional σ . Have students explain why only the first visual change improves the simple resolving-power reference.

- **12–15** Use the constrained third challenge. Ask what would need to change if more length alone cannot separate the pair: the timing spread or another part of a real instrument, not simply the slider.

RESOLUTION DISCUSSION

The app labels each peak's σ and uses Gaussian peak area to conserve packet counts. For narrow peaks, its mass-resolution reference is $m/\Delta m \approx t/(2 \cdot \text{FWHM})$, where $\text{FWHM} = 2.355\sigma$. With a fixed absolute σ , doubling L doubles t while FWHM stays fixed, so R doubles. With a fixed fractional σ , doubling L doubles both values, leaving R unchanged. This is why the third challenge cannot be won by universally maximizing length.

PROMPT BANK

1. What quantity does this ideal TOF clock measure directly, and what quantity is inferred?
2. Why do 200 Da, $z = 1$ and 400 Da, $z = 2$ land together here?
3. In which width convention does a longer tube help? What assumption makes that true?
4. Peak C becomes shorter when σ grows. Why does that not mean fewer ions arrived?
5. Which missing instrument effects might a real analytical lab need before identifying an unknown?

INSTRUCTOR CONTROLS

? reopens the how-to. ⚙ opens presentation mode, presenter's notes, and reset. The same notes appear in the app so a presenter can use this guide before class and the short sequence during class. The phone layout stacks controls under the instrument; no hover-only interaction is required.

SOURCES AND ATTRIBUTION

Iowa State University Chemical Instrumentation Facility, [Mass Spectrometry Tutorial](#), especially its linear TOF section: qV acceleration, field-free drift, and detector timing. Shimadzu, [MALDI basics](#), for pulsed-ionization course context. Sources support the concepts only; they do not endorse this demo. The Morris project record at [Murray State Digital Commons](#) supports a plausible course fit, not a claim that this is its instrument or assignment.